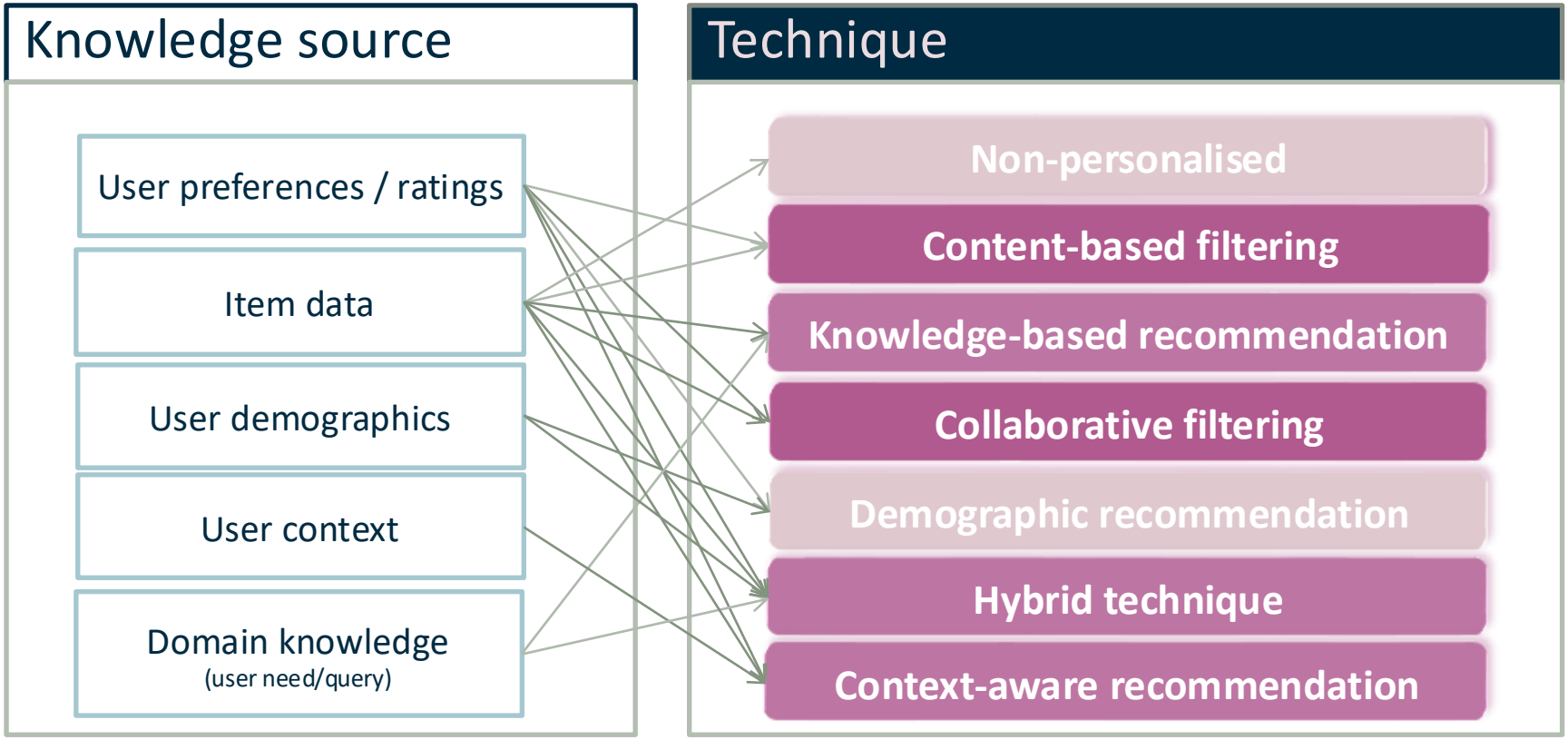
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Content-based Filtering

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Recommendation Techniques



Source: Konstan (2019), Burke (2007)

Lecture outline

Topics	Activities
Defining CBF	Task: example CBF implementation
Applications	Task: computing a Vector Space Model
Benefits, drawbacks	Task: explore datasets, frameworks
Architecture	
Content analysis	
Profile learning	
Filtering component	
Advances in CBF	

Definition, applications, benefits, drawbacks

What is CBF?

- **Content-based filtering**
  - analyses **features** of the items / **documents**
  - previously “**rated**” by a user
  - then builds a **model / profile** of user interests

- **Aim**
  - recommend items similar to the items this user has “liked” in the past

Source: Lops (2011)

Content-Based Movie Recommendation System Using Genre Correlation, Reddy (2019)

- Features considered
  - genres user might prefer
- Approach
  - content-based filtering using genre correlation
- Dataset
  - Movie Lens

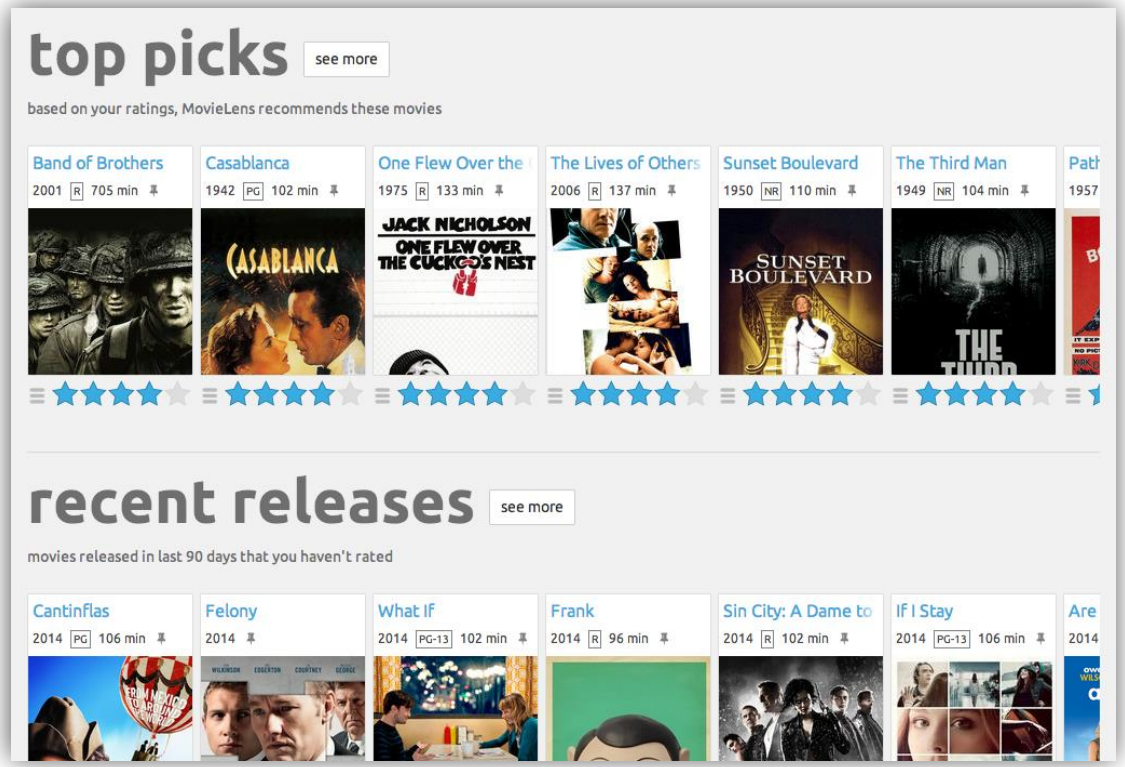


Image source: https://movielens.org/

Publication Recommender System, Wang (2018)

- Finding relevant CS publication venues
  - 66 venues
  - 5 digital libraries (Springer, IEEE, ACM, AAAI and SIAM)
- Content-based filtering model
  - Feature selection model -> chi-square
  - Softmax regression model
- Data
  - abstract or whole manuscript

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Search
Showing results 1 - 30 of 30

Title	Source	Rank
Algorithmica: an international journal in computer science	ERA2010	A*
Journal of Computer and System Sciences	ERA2010	A*
Theoretical Computer Science	ERA2010	A
Science of Computer Programming	ERA2010	A
Computer Science Education	ERA2010	A
Logical Methods in Computer Science	ERA2010	A
Discrete Mathematics and Theoretical Computer Science	ERA2010	B
Journal of Computer Science and Technology	ERA2010	B
International Journal of Foundations of Computer Science	ERA2010	B
Innovations in Teaching and Learning in Information and Computer Sciences	ERA2010	B
Social Science Computer Review	ERA2010	B
International Journal of Applied Mathematics and Computer Sciences	ERA2010	C
Journal of Universal Computer Science	ERA2010	C
Mathematics in Computer Science	ERA2010	C
Egyptian Computer Science Journal	ERA2010	C
IAENG International Journal of Computer Science	ERA2010	C
International Journal of Computer and Information Science and Engineering	ERA2010	C
Automatic Control and Computer Sciences	ERA2010	C
International Journal of Computer Science and Engineering	ERA2010	C



Benefits of CBF

User independence

- Use only ratings to build user profile
- vector space models

Transparency

- Recommendations can be explained
- List content features / descriptions

New Item

- New items recommended
- Not susceptible to first-rater problem

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Source: Lops (2011)

Drawbacks of CBF

Limited Content Analysis

- Limit in the number and type of features
- Domain knowledge needed
- Enough information required to discriminate between P and N items
- Harder to find complements than substitutes
- Weighing attributes and ratings; content interdependency

Over-specialisation

- No inherent method for finding unexpected items
- Lack of serendipity
- Limited novelty

New User

- Need to collect enough ratings
- Recommendations for new user not reliable

User Profile Updating

- throw away/recompute
- mix in new rating; decay old profile over time

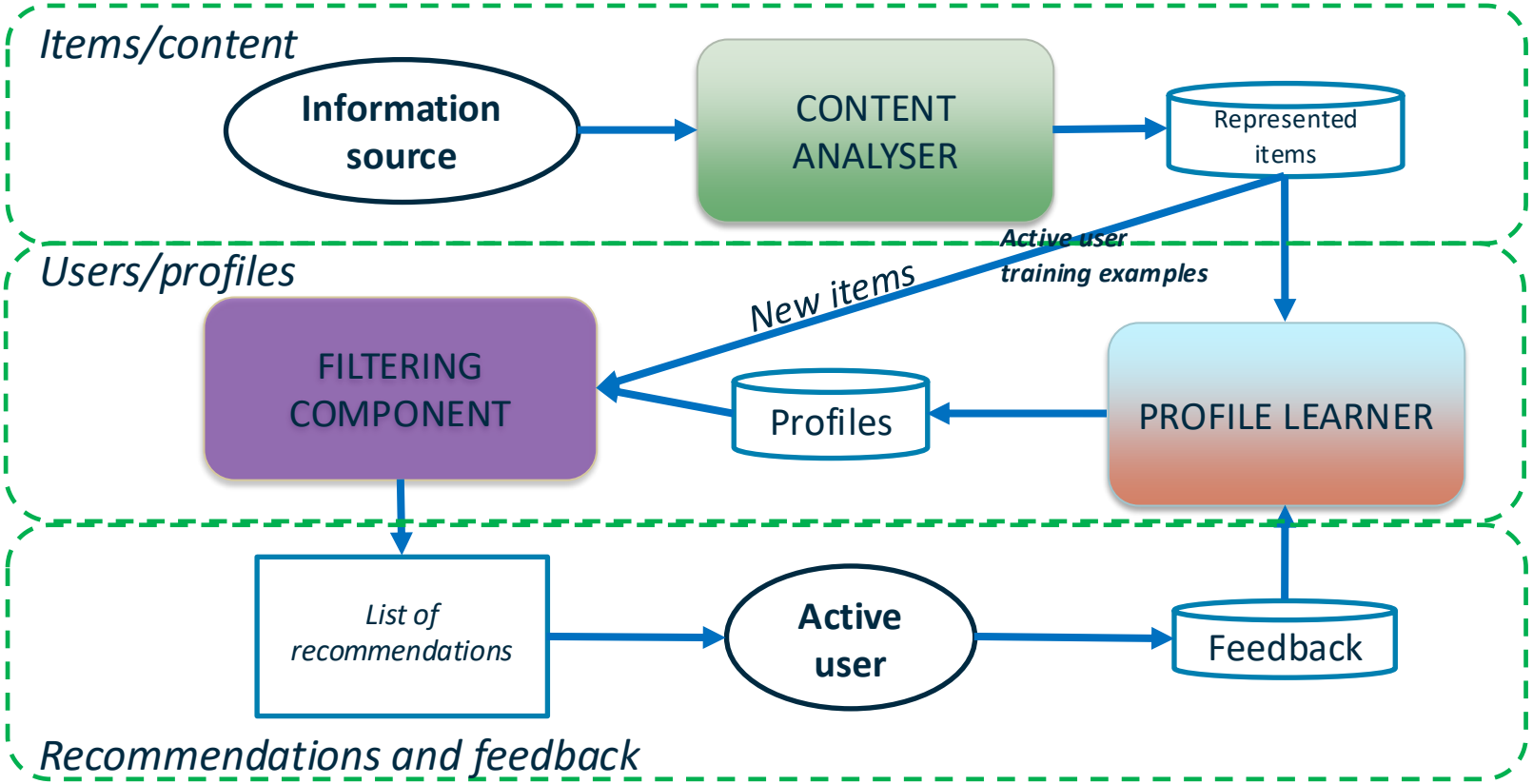
Source: Lops (2011), Konstan (2019)

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CBF Architecture

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CBF Architecture



Source: Lops (2011)

Content analyser

- **Aim**
  - Represent items' content in a structured form
    - Type 1: structured - same number of attributes, with known set of values
    - Type 2: unstructured data *(e.g., tags, posts, opinions, etc.)*

- **Feature extraction**
  - Transforming original information space to target one
    - e.g., Web page -> keywords

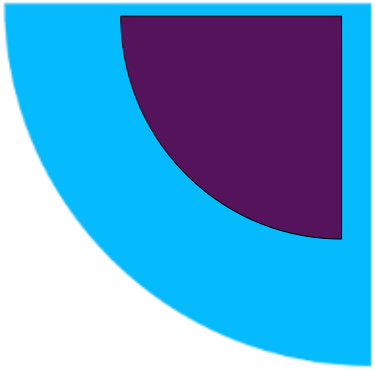
Source: Lops (2011)

Profile learner

- **Aim**:
  - Collect and generalise user preference data
  - construct **user profile**
- **Represented items** repository
- **Feedback** repository
  - Explicit or implicit user interests
  - Inferred from reactions to recommendations
- **User profile**
  - **Vector profile**: aggregate features of preferred items.
  - **Learned model**: predict like probability or a rating from features.

Source: Lops (2011)

Filtering component



Recommendation-list construction:

1. Select eligible candidate items.
2. Score each candidate with the user's profile/model.
3. Sort scores and return the top K items.
4. Collect feedback and update the profile.

Exclude consumed items when appropriate for the domain.

Source: Lops (2011)

Content analysis

Item description

- Structured data
  - Attributes
- Unstructured data
  - **Keyword-based approach**
    - Issues / drawbacks
      - Requires training sets with a large number of examples
      - Lack of “intelligence” - polysemy, synonymy
  - **Semantic analysis**
    - knowledge bases (lexicons or ontologies)
      - “semantic” interpretation of user information needs

Source: Lops (2011), Wang (2018)

Pipeline: Feature extraction, selection and weight

1. **Pre-processing**
   - **Tokenization**
     - Extracting features (tokens, words, grams, terms...)
     - Word vs sentence vs character vs etc.
   - stopwords removal, stemming, lemmatization, lowercase, spellcheck
2. Feature **selection** metric
   - Sort metric values
     - pick **rich informative features**
3. Combine all feature **vectors**
   - remove duplicate terms
   - generate **new feature vector space**
     - all documents represented by vectors of equal length
4. **Weighting** scheme

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Source: Lops (2011), Wang (2018)

Feature selection metrics

Chi-square (χ2)

$$\chi^2(t, c) = \frac{N \times (AD - BC)^2}{(A + C) \times (B + D) \times (A + B) \times (C + D)}$$

Information gain

$$IG(t, c) = Entropy(S) + \frac{A + B}{N_1 + N_2}\left(\frac{A}{A + B}log\left(\frac{A}{A + B}\right) + \frac{B}{A + B}log\left(\frac{B}{A + B}\right)\right) + \frac{C + D}{N_1 + N_2}\left(\frac{C}{C + D}log\left(\frac{C}{C + D}\right) + \frac{D}{C + D}log\left(\frac{D}{C + D}\right)\right)$$

$$Entropy(S) = -\frac{N_1}{N_1 + N_2}log\left(\frac{N_1}{N_1 + N_2}\right) - \frac{N_2}{N_1 + N_2}log\left(\frac{N_2}{N_1 + N_2}\right)$$

Mutual information

$$MI(t, c) = log\left(\frac{A/(A + C)}{(A + B)/N}\right)$$

- A - number of documents including term t, which belongs to category c
- B - number of documents including t, which does not belong to c
- C - number of documents in category c, which does not include t
- D - number of documents in other categories and without term t
- N – size of the corpus; total number of documents

Source: Wang (2018)

Keyword vector

- **Item/document representation**
  - All items represented by vectors of equal length (same terms/dimensions)
  - A vector of term weights
    - in an *n*-dimensional space
    - **dimension** is a term from the dictionary (set of all the words) of the corpus (all items)
      - E.g., in movies these could be: different genres, all possible actors and directors
    - **weight** is the degree of association between the item and the term

	present	upgrade	colour
item1	1	0	1
item2	0	1	1
item3	0	0	0

Source: Konstan (2019)

Weighting scheme: TF-IDF

- Very common conventional weighting scheme
- Terms/features with **higher TF-IDF** are more important
  - rare terms *t* (across documents/items) are not less relevant than frequent terms (IDF) - number of all documents (*N*) with term k (n_k)
  - multiple occurrences of a term *t* in a document *d* is not less relevant than single occurrence (TF)
  - long documents are not preferred (**normalisation**)

$$\text{TF-IDF}(t_k, d_j) = \underbrace{\text{TF}(t_k, d_j)}_{\text{TF}} \cdot \underbrace{log\frac{N}{n_k}}_{\text{IDF}}$$

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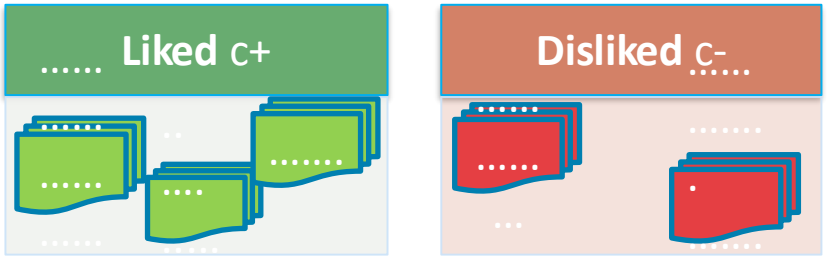
Source: Lops (2011)

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Profiling and filtering

User interaction data

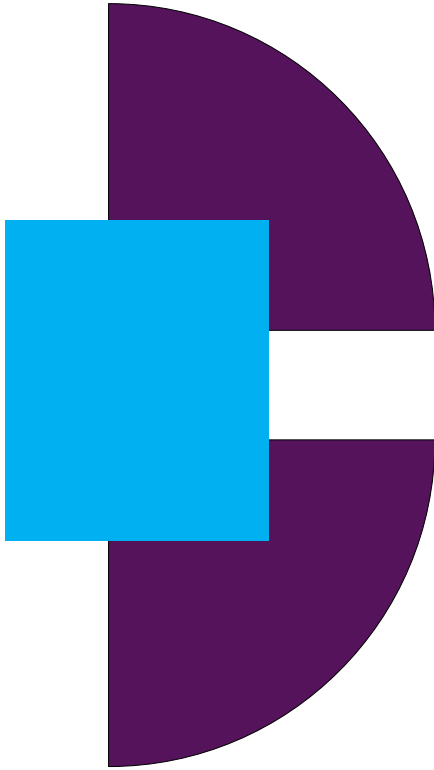
- Training set
  - **Labelled items** based on active user's preferences, e.g. as:
    - Ratings 1-5
    - Liked vs disliked items (classification?)
    - Implicit feedback: viewed vs not viewed items, etc.



Source: Lops (2011)

Methods: Prediction/classification

- **Machine learning** techniques
  - **Supervised learning** algorithms
    - learned from **training set**
    - features extracted in content analysis



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Source: Lops (2011)

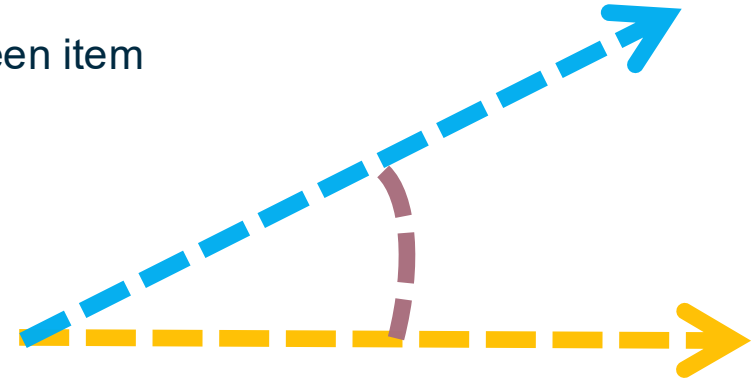
Nearest neighbour methods

- Lazy learners
- Process:
  1. Compare new item to **all** stored items
     - use similarity function
  2. Find k-nearest neighbours
  3. Assign neighbours' class to the new item

Source: Lops (2011)

Comparison/similarity between vectors

- Match two vectors of the **same** length
  1. User profile vector
     - I.e. vector representation of a seen item
  2. Item representation vector
     - of the unseen item



- By applying a distance/**similarity measure**

Source: Amatriain (2011), Lops (2011)

Similarity / distance measures

Cosine	$cos(x,y) = \frac{(x \bullet y)}{\|\|x\|\|\|\|y\|\|}$ $\quad sim(d_i,d_j) = \frac{\sum_k w_{ki} \cdot w_{kj}}{\sqrt{\sum_k w_{ki}^2} \cdot \sqrt{\sum_k w_{kj}^2}}$
Pearson correlation	$Pearson(x,y) = \frac{\Sigma(x,y)}{\sigma_x \times \sigma_y}$
Euclidean	$d(x,y) = \sqrt{\sum_{k=1}^{n}(x_k - y_k)^2}$
Minkowski	$d(x,y) = (\sum_{k=1}^{n} \|x_k - y_k\|^r)^{\frac{1}{r}}$
Mahalanobis	$d(x,y) = \sqrt{(x-y)\sigma^{-1}(x-y)^T}$
Jaccard, Simple Matching	$JC = \frac{M11}{M01+M10+M11}$ $\quad SMC = \frac{number of matches}{number of attributes}$

Source: Amatriain (2011)

Predicting unseen item's class/rating

1. Find k-nearest neighbours
2. Assign neighbours' class to the new item
   - Class of the most similar vector is assigned to the new item
   - Other approaches?

Source: Amatriain (2011), Lops (2011)

Other conventional methods

Decision trees

- **nodes** - labelled by terms
- **Branches** - labelled by term weight
- **leaves** - labelled by categories
- learned by recursively partitioning training data into subgroups

Decision rule

- from all the **rules** that correctly classify all the training examples
  - select the "**best**" one according to a minimality criterion

Logistic (Softmax) regression

- estimates a probability of $p(y = j\ x)$ for input x
  - For each value of $j = 1, \cdots, k$
- output - k dimensional vector
  - k estimated probabilities summing up to 1

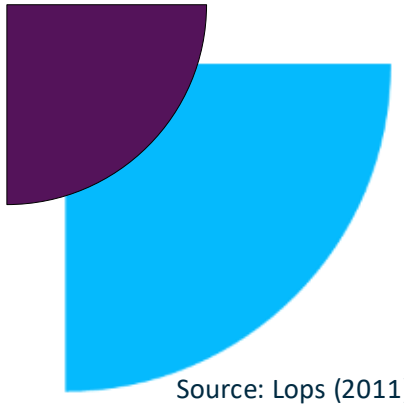
Source: Lops (2011), Wang (2018)

Probabilistic Methods: Naïve Bayes

- NB model
  - $P(c|d)$ a **posteriori** probability of document d belonging to class c
  - $P(c)$ - **a priori** probability
  - Select the class with the highest probability

$$P(c|d) = \frac{P(c)P(d|c)}{P(d)}$$

$$P(c_j|d_i) = P(c_j) \prod_{w \in V_{d_i}} P(t_k|c_j)^{N(d_i,t_k)}$$



Source: Lops (2011)

Advances in CBF

Advances in content analysis

Content-derived representations

- **Word2Vec** on item text: aggregate word embeddings.
- **Sentence-BERT**: encode descriptions into semantic vectors.

Interaction-derived representations

- **Item2Vec** learns from item co-occurrence.
- **BERT4Rec** models sequences of item interactions.

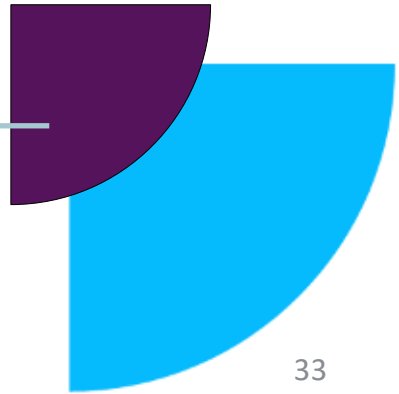


Sources: Reimers & Gurevych (2019); Barkan & Koenigstein (2016); Sun et al. (2019)

Alternative embedding approaches

A semantic-content pipeline

1. Combine useful movie fields into a text description.
2. Encode each description with a **Sentence Transformer**.
3. Aggregate embeddings of preferred items into a profile.
4. Retrieve candidates and rank by cosine similarity.



Modern directions in content recommendations

Multimodal content

- Combine text with image or audio features to describe items.

LLM-assisted representations

- Extract attributes or encode descriptions for new-item matching.
- Verify generated attributes and evaluate latency and bias.



Source: Zhang (2025)

Key topics to take away

- CBF definition
- Benefits / drawbacks

- Architecture
  - Content analyser
  - Profile learner
  - Filtering component

- Vector Space Model
  - TFIDF weighting
- User profile learning
  - Nearest neighbour algorithms
    - Similarity measures: cosine, Pearson correlation, Euclidean, etc.
  - Probabilistic methods

References and reading material

- Ahn, J., Brusilovsky, P., Grady, J., He, D., Syn, S.Y.: Open User Profiles for Adaptive News Systems: Help or Harm? In: C.L. Williamson, M.E. Zurko, P.F. Patel-Schneider, P.J. Shenoy (eds.) Proceedings of the 16th International Conference on World Wide Web, pp. 11–20. ACM (2007)
- Amatriain, X., Jaimes, A., Oliver, N., & Pujol, J. M. (2011). Data mining methods for recommender systems. In *Recommender systems handbook* (pp. 39-71). Springer, Boston, MA.
- Huang, R. (2023). Improved content recommendation algorithm integrating semantic information. Journal of Big Data, 10(1), 84.
- Konstan, J. & Ekstrand, M. (2019). Introduction to Recommender Systems: Non-Personalized and Content-Based. Available: https://www.coursera.org/learn/recommender-systems-introduction/home/welcome
- Lops, P., De Gemmis, M., & Semeraro, G. (2011). Content-based recommender systems: State of the art and trends. In *Recommender systems handbook* (pp. 73-105). Springer, Boston, MA.
- Lops, P., Jannach, D., Musto, C., Bogers, T., & Koolen, M. (2019). Trends in content-based recommendation. *User Modeling and User-Adapted Interaction*, *29*(2), 239-249.
- Reddy, S. R. S., Nalluri, S., Kunisetti, S., Ashok, S., & Venkatesh, B. (2019). Content-Based Movie Recommendation System Using Genre Correlation. In *Smart Intelligent Computing and Applications* (pp. 391-397). Springer, Singapore.
- Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., & Jiang, P. (2019, November). BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In Proceedings of the 28th ACM international conference on information and knowledge management (pp. 1441-1450).
- Wang, D., Liang, Y., Xu, D., Feng, X., & Guan, R. (2018). A content-based recommender system for computer science publications. *Knowledge-Based Systems*, *157*, 1-9.
- Zhang, W., Bei, Y., Yang, L., Zou, H. P., Zhou, P., Liu, A., ... & Yu, P. S. (2025). Cold-start recommendation towards the era of large language models (llms): A comprehensive survey and roadmap. *arXiv preprint arXiv:2501.01945*.